Lesson 3 · Build a rain gauge · for your pupils

Make a bottle rain gauge

40 minutes · Ages 10–16 · adapt for other ages below · No electricity needed · No devices needed · Works outdoors

Big question

How can we measure rain with a number?

Pupils will be able to...

- Explain that rainfall is measured in millimetres of depth.
- Build a rain gauge from a bottle, with a zero line, a millimetre scale and a funnel.
- Choose a good place for the gauge and agree a daily reading routine.

You need

- One clear plastic bottle with straight, smooth sides per group
- Scissors or a knife for the teacher only; tape
- Small stones, water, a ruler and a marker or pen
- Paper strips for a millimetre scale

Before class

Before class, cut the top off each bottle where the sides start to curve, and tape the cut edges. Check that each bottle's sides are straight. Walk the school grounds and pick two or three open spots to discuss.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Ask: "Did it rain a lot or a little last time?" Let pupils disagree. Write their words on the board — "a lot", "a bit", "heavy".	*"Words like "a lot" mean different things to different people. Today we'll measure rain with a number."*	Describe the last rain and notice that their words disagree.
Teach 5–12 min	Explain millimetres of rain: the depth on flat ground. Show a wide and a narrow straight-sided container and ask: in the same rain, which fills deeper? Explain that both reach the same depth.	*"A wide bottle catches more water, but the depth is the same. Rainfall is depth."*	Predict which container fills deeper, then explain why both match.
Build 12–27 min	Groups add stones and water just to cover them, mark the 0 line, and tape on a paper scale marked every millimetre from a ruler. Then they turn the cut top into the bottle as a funnel and tape it.	*"Is your 0 line level? Does every millimetre on your scale match the ruler?"*	Build the gauge; check each other's scales with a ruler.
Place it 27–35 min	Walk to the spots you chose. Groups judge each one: open sky? Far from roofs and trees? Safe from feet and animals? Place the gauge in the best spot.	*"Look up. Is there anything above the gauge that could block rain or drip water in?"*	Compare spots, give reasons and choose one together.
Wrap-up 35–40 min	Back in class, draw a reading rota and a table on the board: date, time, rain (mm). Exit ticket: one rule for reading the gauge.	*"Same time every day, eye level, write it down, then pour it back to 0."*	Sign up for the rota; write the exit ticket.

Check understanding
Ask each group: “Why straight sides?” and “Why not under the tree?” Listen for “every millimetre equal” and “leaves block rain”. Check that scales start at the 0 line, not the bottom of the bottle.

Exit ticket
One rule for reading the rain gauge, and why it matters.

Common mix-ups

Pupils may think…	Help them see…
A bigger bottle measures more rain.	It catches more water, but over a wider opening, so the water stands at the same depth. Rainfall is depth, not amount.
A shady tree is a good spot.	Leaves block rain and drip it elsewhere. Pick open ground, well away from trees and roofs.
We can leave the water in for tomorrow.	Then tomorrow’s reading would include today’s rain. Write the reading, then pour the gauge back down to the 0 line.

Adapt it

Younger pupils
Ages 6–9: you build the gauge; pupils add stones and help tape the scale. Each day, one pupil colours a paper strip up to the water level.

Older pupils and university
Groups build two gauges — one straight-sided, one that narrows — and compare readings after rain. Discuss why weather stations use a funnel and a narrow measuring tube.

Large classes
Build one gauge at the front with volunteers while rows check each step. Give each row a day of the week to read it.

No board or no space
Do the lesson outside. Keep the reading table on a large sheet pinned to the classroom door, or in a notebook the reader carries.

Short on time? Build one gauge as a class demonstration, place it together and set up the reading rota (20 minutes).

What you need to know

A rain gauge collects rain so you can measure it in millimetres — how deep it would stand on flat ground. A straight-sided bottle, a funnel and a careful routine are enough.

Rain gauge
A container that collects rain so you can measure how much fell. A simple one can be made from a plastic bottle.

Millimetre (mm)
One thousandth of a metre — the smallest marks on a school ruler. Rainfall is measured in millimetres of depth.

Funnel
A cone that guides liquid into a container. Here, the bottle’s own top, turned upside down, guides rain in and keeps most leaves out.

Zero line
The level you measure from. If the bottle’s bottom is bumpy, add stones and water to cover it and mark that level 0.

My notes — how did it go?